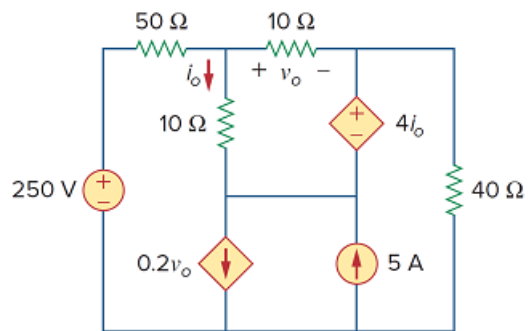


Problem

Find v_o and i_o in the circuit of Fig. 3.108.

Figure 3.108:



Step-by-step solution

Step 1 of 5

Consider the following circuit diagram:

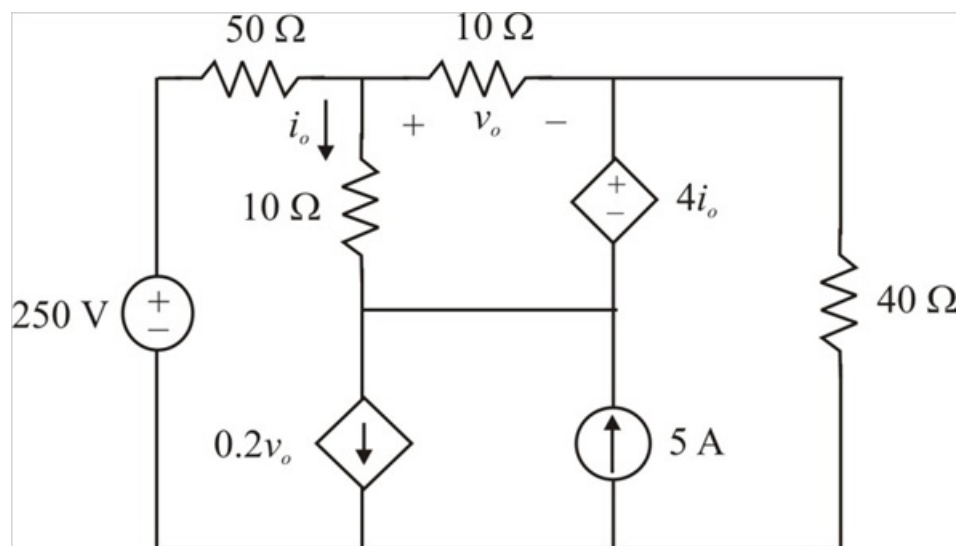


Figure 1

Step 2 of 5

Label the circuit as shown in Figure 2:

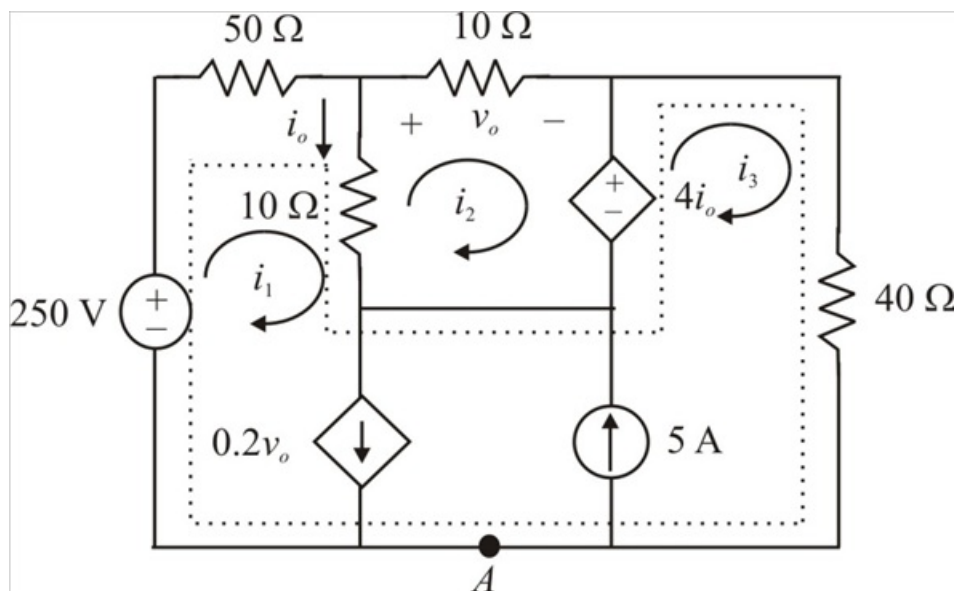


Figure 2

Step 3 of 5

The following is the expression for the current i_o from the circuit in Figure 2.

$$i_o = i_1 - i_2$$

The following is the expression for the voltage v_o from the circuit in Figure 2.

$$v_o = (10 \Omega) i_2$$

Kirchhoff's voltage law states that the algebraic sum of all voltages around a closed path (or loop) is zero.

Apply Kirchhoff's voltage law to the loop 2 of circuit in Figure 2 to get the following expression:

$$\begin{aligned} (10 \Omega)(i_2 - i_1) + (10 \Omega)i_2 + 4i_o &= 0 \\ (10 \Omega)(i_2 - i_1) + (10 \Omega)i_2 + 4(i_1 - i_2) &= 0 \quad (\text{since, } i_o = i_1 - i_2) \\ (10 \Omega)i_2 - (10 \Omega)i_1 + (10 \Omega)i_2 + 4i_1 - 4i_2 &= 0 \\ -6i_1 + 16i_2 &= 0 \\ -3i_1 + 8i_2 &= 0 \quad \dots\dots (1) \end{aligned}$$

The meshes 1 and 3 form a super mesh. Apply Kirchhoff's voltage law to the super mesh.

$$\begin{aligned} -250 \text{ V} + (50 \Omega)i_1 + (10 \Omega)i_o - 4i_o + (40 \Omega)i_3 &= 0 \\ -250 \text{ V} + (50 \Omega)i_1 + (10 \Omega)(i_1 - i_2) - 4(i_1 - i_2) + (40 \Omega)i_3 &= 0 \\ -250 \text{ V} + (50 \Omega)i_1 + (10 \Omega)i_1 - (10 \Omega)i_2 - 4i_1 + 4i_2 + (40 \Omega)i_3 &= 0 \\ 56i_1 - 6i_2 + 40i_3 &= 250 \\ 28i_1 - 3i_2 + 20i_3 &= 125 \quad \dots\dots (2) \end{aligned}$$

Step 4 of 5

Apply Kirchhoff's current law at node A of circuit in Figure 2.

$$0.2v_o + i_3 = i_1 + 5 \text{ A}$$

$$0.2(10i_2) + i_3 = i_1 + 5 \text{ A} \quad \left[\text{since, } v_o = (10 \Omega)i_2 \right]$$

$$2i_2 + i_3 = i_1 + 5 \text{ A}$$

$$-i_1 + 2i_2 + i_3 = 5 \dots\dots (3)$$

Multiply equation (3) by the factor (20) and subtract the result from equation (2).

$$28i_1 - 3i_2 + 20i_3 = 125$$

$$\frac{(20)(-i_1 + 2i_2 + i_3) = (20)(5)}{28i_1 - 3i_2 + 20i_3 = 125}$$

$$\frac{(-) - 20i_1 + 40i_2 + 20i_3 = 100}{48i_1 - 43i_2 + 0 = 25}$$

Therefore,

$$48i_1 - 43i_2 = 25 \dots\dots (4)$$

Multiply equation (1) by the factor (16) and add equation (4) to the result.

$$(16)(-3i_1 + 8i_2) = 0$$

$$\frac{48i_1 - 43i_2 = 25}{-48i_1 + 128i_2 = 0}$$

$$\frac{48i_1 - 43i_2 = 25}{0 + 85i_2 = 25}$$

Simplify the expression to calculate i_2 .

$$\begin{aligned} i_2 &= \frac{25}{85} \\ &= 0.294 \text{ A} \end{aligned}$$

Calculate the voltage v_o :

$$\begin{aligned} v_o &= (10 \Omega)i_2 \\ &= 10(0.294 \text{ A}) \\ &= 2.94 \text{ V} \end{aligned}$$

Therefore, the voltage v_o is $\boxed{2.94 \text{ V}}$.

Step 5 of 5

Substitute 0.294 A for i_2 in equation (1) to calculate the current i_1 .

$$-3i_1 + 8i_2 = 0$$

$$-3i_1 + 8(0.294) = 0$$

$$-3i_1 = -2.353$$

$$\begin{aligned} i_1 &= \frac{-2.353}{-3} \\ &= 0.784 \text{ A} \end{aligned}$$

Calculate the current i_o :

$$\begin{aligned} i_o &= i_1 - i_2 \\ &= 0.784 \text{ A} - 0.294 \text{ A} \\ &= 0.49 \text{ A} \\ &= 490 \text{ mA} \end{aligned}$$

Therefore, the current i_o is $\boxed{490 \text{ mA}}$.